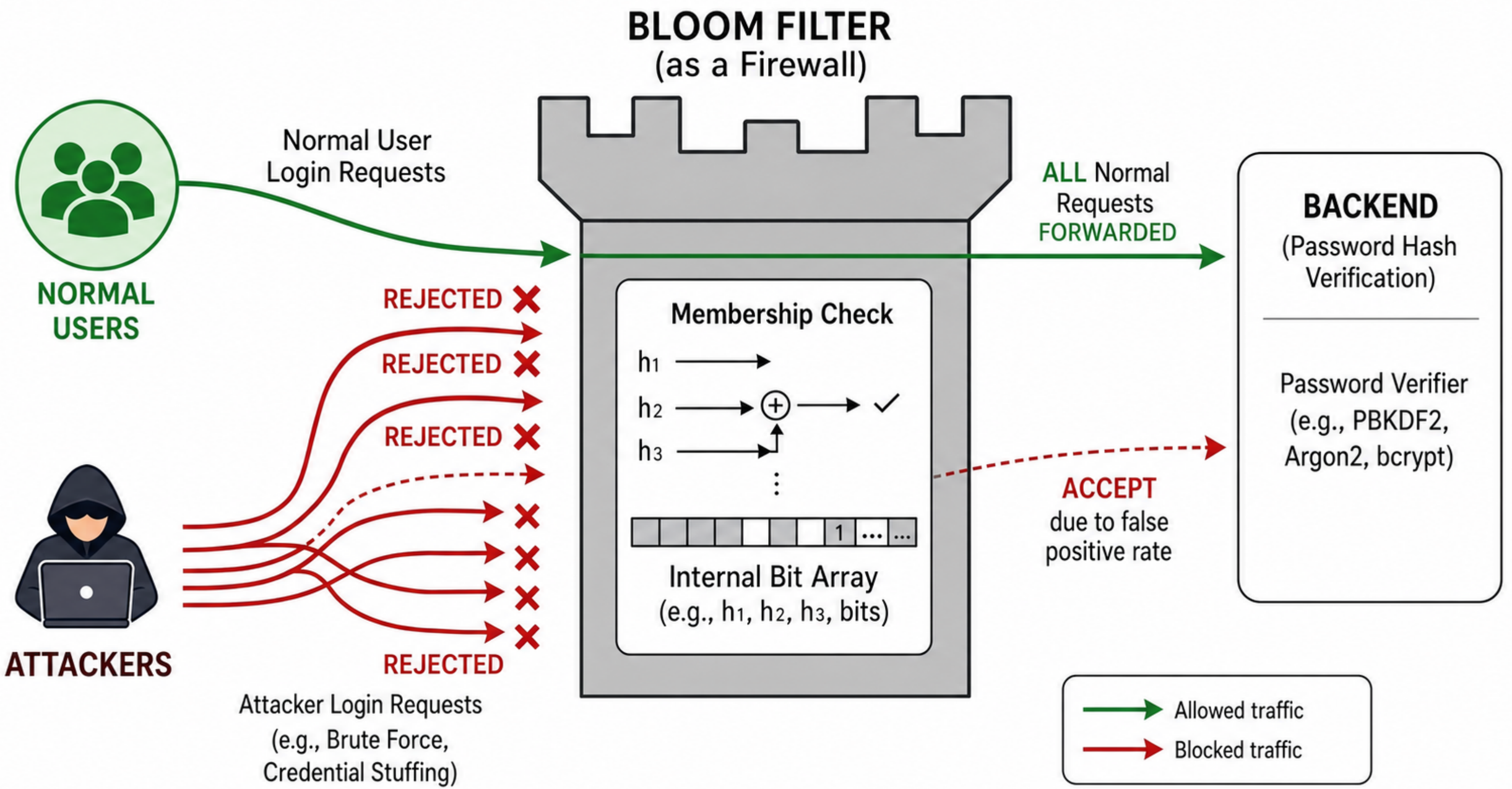

BLOOM FILTER
(as a Firewall)
NORMAL USERS
Normal User Login Requests
ALL Normal Requests FORWARDED
BACKEND
(Password Hash Verification)
Password Verifier (e.g., PBKDF2, Argon2, bcrypt)
Membership Check
h1
h2
h3
1
Internal Bit Array
(e.g., h1, h2, h3, bits)
ACCEPT
due to false positive rate
REJECTED
REJECTED
REJECTED
REJECTED
ATTACKERS
Attacker Login Requests
(e.g., Brute Force, Credential Stuffing)
Allowed traffic
Blocked traffic